

2

2



- Recall that the round trip rate is a symmetric function of the rejection rates $r_n = r(\beta_{n-1}, \beta_n)$

$$\tau(r_1, \dots, r_N) = \frac{1}{2 + 2 \sum_{n=1}^N \frac{r_n}{1 - r_n}}$$

► Since $r_1 + \dots + r_N \approx \Lambda$ when N is large enough have the optimal round trip rate satisfies:

$$\text{maximize :} \quad \tau(r_1, \dots, r_N)$$

$$\text{Constraint :} \quad \sum_{n=1}^N r_n = \Lambda, \quad r_n \geq 0$$

► **Solution:** Round-trip rate is maximised when r_n is constant in n satisfying

Since $r(\beta_{n-1}, \beta_n)$ the integral of $\lambda(\beta)$ from β_{n-1} to β_n the optimal schedule satisfies,

$$\Lambda(\beta_n) = \Lambda \frac{n}{N}$$

$$r(\beta_{n-1}, \beta_n) \approx \frac{\Lambda}{N}$$

Optimal schedule achieves constant length travel path with constant speed.

OPTIMAL SCHEDULE

- ▶ Recall that the round trip rate is a symmetric function of the rejection rates $r_n = r(\beta_{n-1}, \beta_n)$

$$\tau(r_1, \dots, r_N) = \frac{1}{2 + 2 \sum_{n=1}^N \frac{r_n}{1 - r_n}}$$

- ▶ Since $r_1 + \dots + r_N \approx \Lambda$ when N is large enough have the optimal round trip rate satisfies:

$$\text{maximize : } \tau(r_1, \dots, r_N)$$

$$\text{Constraint : } \sum_{n=1}^N r_n = \Lambda, \quad r_n \geq 0$$

- ▶ **Solution:** Round-trip rate is maximised when r_n is constant in n satisfying

$$r(\beta_{n-1}, \beta_n) \approx \frac{\Lambda}{N}$$

- ▶ Since $r(\beta_{n-1}, \beta_n)$ the integral of $\lambda(\beta)$ from β_{n-1} to β_n the optimal schedule satisfies,

$$\Lambda(\beta_n) = \Lambda \frac{n}{N}$$

- ▶ Optimal schedule accumulates constant length traverses path with constant speed.

EXAMPLE: ISING MODEL ($D = 400$)

